

## **INTERFACE (ABSTRACTION) TASK: -**

List of tasks related to Interface (Abstraction)–

Question 1. What is abstraction? How can we achieve an abstraction?

Question 2. What is interface. Write the syntax rules for it?

Question 3. Difference between Class and Interface in java?

Question 4. Can interface support multiple inheritance in java?

Question 5. How can we achieve interface in java?

Question 6. Why should we need to initialize the variable of interface mandatory?



**Question 7. Check the declaration of interface variable is it correct or not.**

**public interface I**

```
{  
    int x=10;  
    public int y;  
    static z=10;  
    final int p=10;  
    public static int Q=10;  
    public final int R;  
    static final int S=10;  
    public static final int T=10;  
}
```

**Question 8. Check the declaration of interface method is it correct or not.**

**public interface I**

```
{
```



```
public void m1();  
void m2();  
abstract void m3();  
public abstract void m4()  
abstract public void m5();  
m6();  
}
```

**Question 9. Check the extends and implements of Class and Interface is it correct or not.**

- 1) public class Class1 implements I1
- 2) public class Class1 implements I1, I2, I3
- 3) public class Class2 extends Class3, Class1
- 4) public class Class3 extends Class1 implements I1, I2, I3
- 5) public class Class3 implements I1 extends Class1
- 6) public class Class2 extends Class1
- 7) public interface I1 implements I2
- 8) public interface I2 extends I1, I3



**CJC**  
Complete Java Classes

**Question 10. Solve the below program and get the output?**

**interface printable**

```
{  
void print ();  
}
```

**class A implements printable**

```
{  
public void print ()  
{  
//print: Hello  
}  
}
```

main ()

```
{  
create object of A using parent to child  
and call print method  
}
```

**Question 11. Solve the below program and get the output?**



**interface Drawable**

```
{  
void draw ();  
}
```

**class Rectangle implements Drawable {**

```
public void draw () {  
//print: drawing rectangle  
}  
}
```

**class Circle implements Drawable {**

```
public void draw ()  
{  
//print: drawing circle  
}  
}
```

**class TestInterface1 {**



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```
main () {  
    Drawable d = new Circle ();  
        Rectangle ();  
    //call circle and rectangle methods.  
    d.draw();  
}  
}
```

**Question 12. Solve the below program and get the output?**

**interface Printable**

```
{  
    void print ();  
}
```

**interface Showable**

```
{  
    void show ();  
}
```

**class A implements Printable, Showable {**



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Complete Java Classes

```
public void print () {  
    //print: Hello  
}  
  
public void show () {  
    //print: Welcome  
}  
  
main () {  
    Printable obj = new A();  
    Showable obj1 = new A();  
    //call all methods  
}  
}
```

**Question 13. Solve the below program and get the output?**

```
interface Printable1 {  
  
    void print ();  
  
}  
  
interface Showable1 {
```



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```
void print ();
```

```
}
```

```
class Test implements Printable, Showable
```

```
{
```

```
public void print () {
```

```
print: Cjc classes
```

```
}
```

```
main () {
```

```
Test t = new Test ();
```

```
t.print ();
```

```
}
```

```
}
```

**Question 14. Solve the below program and get the output?**

```
interface Printable
```

```
{
```

```
void print ();
```

```
}
```





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**interface Showable extends Printable**

```
{  
void show ();  
}
```

**class Test implements Showable**

```
{  
public void print ()  
{  
//print: cjc classes  
}  
public void show ()  
{  
//print: Welcome  
}  
main ()  
{  
Test t = new Test ();
```



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Complete Java Classes

```
t.print ();  
t.show ();  
}  
}
```



**THANK YOU**

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